

Spectral Embedding of Platonic Solids

Tetrahedron

Eigenvalues:

[-1. -1. -1. 3.]

Eigenvectors for $\lambda = -1.0$:

Eigenvector 1:

[0. -0.5774 0.7887 -0.2113]

Eigenvector 2:

[0. -0.5774 -0.2113 0.7887]

Eigenvector 3:

[-0.866 0.2887 0.2887 0.2887]

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Eigenvectors for $\lambda = 3.0$:

Eigenvector 1:

[-0.5 -0.5 -0.5 -0.5]

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Cube

Eigenvalues:

[-3. -1. -1. -1. 1. 1. 1. 3.]

Eigenvectors for $\lambda = -3.0$:

Eigenvector 1:

[0.3536 -0.3536 0.3536 -0.3536 -0.3536 0.3536 -0.3536 0.3536]

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Eigenvectors for $\lambda = -1.0$:

Eigenvector 1:

$[-0.5914 \ 0.0852 \ 0.3391 \ 0.1672 \ 0.3391 \ 0.1672 \ -0.5914 \ 0.0852]$

Eigenvector 2:

$[-0.1587 \ 0.4701 \ -0.4761 \ 0.1647 \ -0.4761 \ 0.1647 \ -0.1587 \ 0.4701]$

Eigenvector 3:

$[0. \ 0.3831 \ 0.1826 \ -0.5656 \ 0.1826 \ -0.5656 \ 0. \ 0.3831]$

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Eigenvectors for $\lambda = 1.0000000000000002$:

Eigenvector 1:

$[-0.0031 \ 0.4864 \ 0.5127 \ 0.0232 \ -0.5127 \ -0.0232 \ 0.0031 \ -0.4864]$

Eigenvector 2:

$[-0.6124 \ -0.2064 \ 0.2013 \ -0.2047 \ -0.2013 \ 0.2047 \ 0.6124 \ 0.2064]$

Eigenvector 3:

$[5.000e-04 \ 3.096e-01 \ -2.675e-01 \ -5.767e-01 \ 2.675e-01 \ 5.767e-01 \ -5.000e-04 \ -3.096e-01]$

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Eigenvectors for $\lambda = 3.00000000000000013$:

Eigenvector 1:

$[0.3536 \ 0.3536 \ 0.3536 \ 0.3536 \ 0.3536 \ 0.3536 \ 0.3536 \ 0.3536]$

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Octahedron

Eigenvalues:

$[-2. \ -2. \ -0. \ -0. \ 0. \ 4.]$

Eigenvectors for $\lambda = -2.0000000000000004$:

Eigenvector 1:

```
[ 4.000e-04 -5.002e-01  4.998e-01 -5.002e-01  4.998e-01  4.000e-04]
```

Eigenvector 2:

```
[ 0.5774 -0.2883 -0.2891 -0.2883 -0.2891  0.5774]
```

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Eigenvectors for $\lambda = -2.993751573463213e-17$:

Eigenvector 1:

```
[-0.6972  0.1163 -0.0195 -0.1163  0.0195  0.6972]
```

Eigenvector 2:

```
[-0.0365 -0.1014  0.6988  0.1014 -0.6988  0.0365]
```

Eigenvector 3:

```
[-0.1121 -0.6901 -0.106  0.6901  0.106  0.1121]
```

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Eigenvectors for $\lambda = 4.0$:

Eigenvector 1:

```
[-0.4082 -0.4082 -0.4082 -0.4082 -0.4082 -0.4082]
```

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Dodecahedron

Eigenvalues:

```
[-2.2361 -2.2361 -2.2361 -2.      -2.      -2.      -2.      -0.      0.
  0.      0.      1.      1.      1.      1.      1.      2.2361  2.2361
  2.2361  3.      ]
```

Eigenvectors for $\lambda = -2.2360679774997894$:

Eigenvector 1:

```
[-0.3873  0.2887 -0.1291 -0.1291  0.2887  0.2887 -0.1291 -0.1291  0.1291
  0.1291 -0.2887  0.1291  0.1291 -0.1291 -0.1291  0.1291  0.1291 -
0.2887
  0.3873 -0.2887]
```

Eigenvector 2:

```
[-0.      0.2528 -0.2373  0.3029 -0.081  -0.1718  0.359  -0.328  0.1217
 -0.0251  0.1718 -0.359  0.328  -0.1217  0.0251  0.2373 -0.3029  0.081]
```

```

0.      -0.2528]

Eigenvector 3:
[ 0.      -0.0524  0.2775 -0.2039  0.2451 -0.1928  0.0668 -0.1604  0.3443
 -0.3643  0.1928 -0.0668  0.1604 -0.3443  0.3643 -0.2775  0.2039 -
0.2451
 -0.      0.0524]

```

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Eigenvectors for $\lambda = -2.0000000000000004$:

```

Eigenvector 1:
[-0.4472  0.2981 -0.0745 -0.0745  0.2981  0.2981 -0.0745 -0.0745 -
0.0745
 -0.0745  0.2981 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745 -0.0745  0.2981
 -0.4472  0.2981]

```

```

Eigenvector 2:
[ 0.      0.3316 -0.3318  0.1363 -0.136  -0.1956  0.1953 -0.3313  0.1357
 0.1958 -0.1956  0.1953 -0.3313  0.1357  0.1958 -0.3318  0.1363 -
0.136
 -0.      0.3316]

```

```

Eigenvector 3:
[-0.      0.0338 -0.0855  0.3511 -0.2994  0.2656 -0.3173  0.018  0.2476
 -0.2138  0.2656 -0.3173  0.018  0.2476 -0.2138 -0.0855  0.3511 -
0.2994
 0.      0.0338]

```

```

Eigenvector 4:
[-0.      0.0065  0.2775 -0.2293 -0.0547  0.0482  0.2358 -0.2905  0.3387
 -0.3322  0.0482  0.2358 -0.2905  0.3387 -0.3322  0.2775 -0.2293 -
0.0547
 0.      0.0065]

```

Applying PCA (dim > 3 → reducing to 3D)

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Eigenvectors for $\lambda = -5.5161000268398e-16$:

```

Eigenvector 1:
[ 2.000e-04  3.383e-01  2.532e-01 -2.552e-01 -3.364e-01 -1.800e-
03

```

```

-8.320e-02 -2.534e-01 -2.551e-01 -8.300e-02 1.800e-03 8.320e-
02
2.534e-01 2.551e-01 8.300e-02 -2.532e-01 2.552e-01 3.364e-
01
-2.000e-04 -3.383e-01]

```

Eigenvector 2:

```

[ 0.4285 0.0526 -0.2679 -0.2866 0.0734 -0.126 -0.1945 -0.1606 0.1419
0.234 0.126 0.1945 0.1606 -0.1419 -0.234 0.2679 0.2866 -
0.0734
-0.4285 -0.0526]

```

Eigenvector 3:

```

[ 0.1222 -0.2364 0.1773 0.1268 -0.1898 0.4263 -0.0126 -0.2995 0.249
0.1097 -0.4263 0.0126 0.2995 -0.249 -0.1097 -0.1773 -0.1268 0.1898
-0.1222 0.2364]

```

Eigenvector 4:

```

[-0.0384 -0.164 0.1808 -0.1914 0.213 -0.049 0.3938 -0.1424 -
0.2298
0.3554 0.049 -0.3938 0.1424 0.2298 -0.3554 -0.1808 0.1914 -
0.213
0.0384 0.164 ]

```

Applying PCA (dim > 3 → reducing to 3D)

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Eigenvectors for = 1.0000000000000002:

Eigenvector 1:

```

[ 0.0005 -0.4371 -0.0814 0.2351 0.2762 0.1614 0.0403 -0.3562 0.0406
0.1206 0.1614 0.0403 -0.3562 0.0406 0.1206 -0.0814 0.2351 0.2762
0.0005 -0.4371]

```

Eigenvector 2:

```

[-0.0171 -0.0122 0.166 -0.2511 -0.3065 0.3017 -0.1106 -0.1611 -
0.0383
0.4293 0.3017 -0.1106 -0.1611 -0.0383 0.4293 0.166 -0.2511 -
0.3065
-0.0171 -0.0122]

```

Eigenvector 3:

```

[-0.4984 -0.1536 0.1923 0.1948 -0.1535 -0.1913 0.1561 0.1526 0.15
0.151 -0.1913 0.1561 0.1526 0.15 0.151 0.1923 0.1948 -
0.1535]

```

-0.4984 -0.1536]

Eigenvector 4:

```
[ -0.0361 -0.1846 -0.4183 -0.275 -0.0117 0.1601 0.1549 0.2699 0.2995
 0.0413 0.1601 0.1549 0.2699 0.2995 0.0413 -0.4183 -0.275 -
0.0117
-0.0361 -0.1846]
```

Eigenvector 5:

```
[ 0.0043 0.0334 0.0621 -0.1345 0.2368 -0.2659 -0.4333 -0.0329 0.367
 0.1631 -0.2659 -0.4333 -0.0329 0.367 0.1631 0.0621 -0.1345 0.2368
 0.0043 0.0334]
```

Applying PCA (dim > 3 → reducing to 3D)

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Eigenvectors for $\lambda = 2.2360679774997902$:

Eigenvector 1:

```
[ 0.0012 0.2585 0.3036 0.0741 -0.1128 -0.1431 0.0251 0.2732 0.3275
 0.3462 0.1431 -0.0251 -0.2732 -0.3275 -0.3462 -0.3036 -0.0741 0.1128
-0.0012 -0.2585]
```

Eigenvector 2:

```
[ -0.3863 -0.2884 -0.1136 -0.1035 -0.2721 -0.3034 -0.1542 -0.145 0.1185
 0.1379 0.3034 0.1542 0.145 -0.1185 -0.1379 0.1136 0.1035 0.2721
 0.3863 0.2884]
```

Eigenvector 3:

```
[ 0.0271 0.0027 0.212 0.3658 0.2515 -0.1936 -0.3544 -0.2331 -
0.1695
 0.1056 0.1936 0.3544 0.2331 0.1695 -0.1056 -0.212 -0.3658 -
0.2515
-0.0271 -0.0027]
```

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Eigenvectors for $\lambda = 3.0$:

Eigenvector 1:

```
[0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236
 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236 0.2236]
```

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Icosahedron

Eigenvalues:

```
[-2.2361 -2.2361 -2.2361 -1.      -1.      -1.      -1.      -1.      2.2361
 2.2361  2.2361  5.      ]
```

Eigenvectors for $\lambda = -2.2360679774997907$:

Eigenvector 1:

```
[ 0.4999 -0.2331 -0.2148 -0.2248 -0.2169  0.2282  0.2248  0.2169 -
0.2282
 0.2331  0.2148 -0.4999]
```

Eigenvector 2:

```
[-2.000e-04  7.420e-02  1.994e-01  4.424e-01 -3.191e-01  3.964e-
01
-4.424e-01  3.191e-01 -3.964e-01 -7.420e-02 -1.994e-01  2.000e-
04]
```

Eigenvector 3:

```
[ 0.0109  0.4361 -0.4051  0.0609 -0.318  -0.2018 -0.0609  0.318  0.2018
-0.4361  0.4051 -0.0109]
```

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Eigenvectors for $\lambda = -1.0000000000000002$:

Eigenvector 1:

```
[ 2.000e-04 -2.919e-01 -4.691e-01  2.831e-01  2.238e-01  2.539e-
01
 2.831e-01  2.238e-01  2.539e-01 -2.919e-01 -4.691e-01  2.000e-
04]
```

Eigenvector 2:

```
[ 0.1028 -0.5464  0.3768 -0.0623 -0.0708  0.1999 -0.0623 -0.0708  0.1999
-0.5464  0.3768  0.1028]
```

Eigenvector 3:

```
[-0.6326  0.0294  0.1955  0.0871  0.1894  0.1312  0.0871  0.1894  0.1312
 0.0294  0.1955 -0.6326]
```

Eigenvector 4:

```
[ 0.0767 -0.1094  0.0201 -0.2927  0.5707 -0.2654 -0.2927  0.5707 -
0.2654
-0.1094  0.0201  0.0767]
```

Eigenvector 5:
[-0.0012 0.1417 -0.1263 -0.4893 0.0013 0.4739 -0.4893 0.0013 0.4739
0.1417 -0.1263 -0.0012]

Applying PCA (dim > 3 → reducing to 3D)

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Eigenvectors for $\lambda = 2.236067977499791$:

Eigenvector 1:
[0.0159 -0.2173 0.3055 -0.0387 -0.4299 -0.4158 0.0387 0.4299 0.4158
0.2173 -0.3055 -0.0159]

Eigenvector 2:
[-0.0321 0.3712 0.3186 -0.4584 -0.109 0.1942 0.4584 0.109 -
0.1942
-0.3712 -0.3186 0.0321]

Eigenvector 3:
[-0.4987 -0.255 -0.235 -0.1959 -0.2308 0.1985 0.1959 0.2308 -
0.1985
0.255 0.235 0.4987]

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Eigenvectors for $\lambda = 5.0$:

Eigenvector 1:
[0.2887 0.2887 0.2887 0.2887 0.2887 0.2887 0.2887 0.2887 0.2887 0.2887
0.2887 0.2887]

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